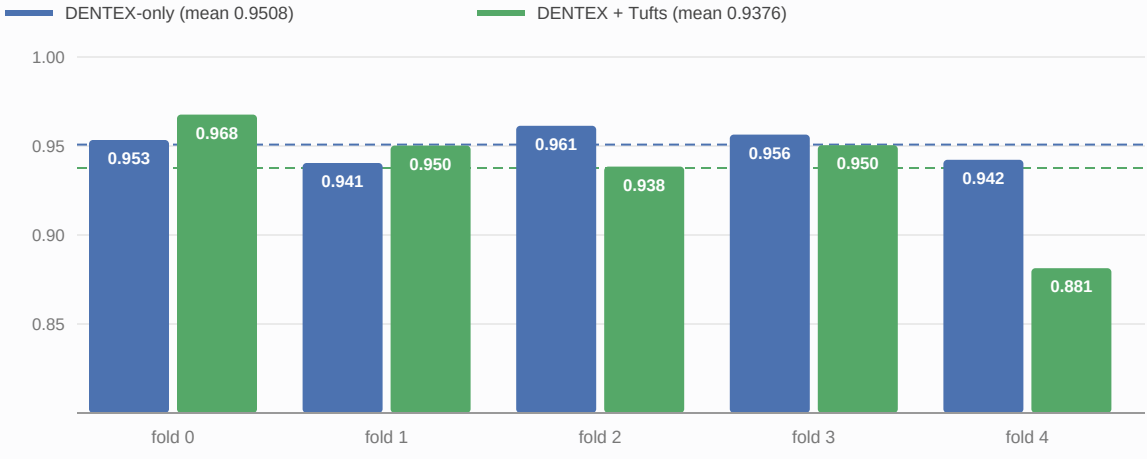
Grounding model: 5-fold CV and held-out verification

YOLOv8m backend of locate_tooth (one class per FDI quadrant-position pair) - target-filtered protocol

in-fold CV: target mAP50 per fold



training pool

DENTEX 1,339 images · 21,624 boxes

Tufts 1,000 images · 25,184 boxes

combined 2,339 images · 46,808 boxes

effect of co-training with Tufts

precision 0.9637 -> 0.9680

raw val mAP50 0.5820 -> 0.8695

(different protocol from the target-filtered CV above - both measured)

held-out official DENTEX validation

50 images · 46 pathological · 182 target boxes

best: DENTEX + Tufts, fold 1

target mAP50 0.9593 · precision 0.9864

deployed as the locate_tooth backend

trace-generation guarantee

when the requested tooth exists in the ground truth, locate_tooth returns its exact box - the detector's measured accuracy governs only the unannotated-tooth fallback; what the model is shown is then perturbed independently (tiered), keeping accept-vs-nudge live